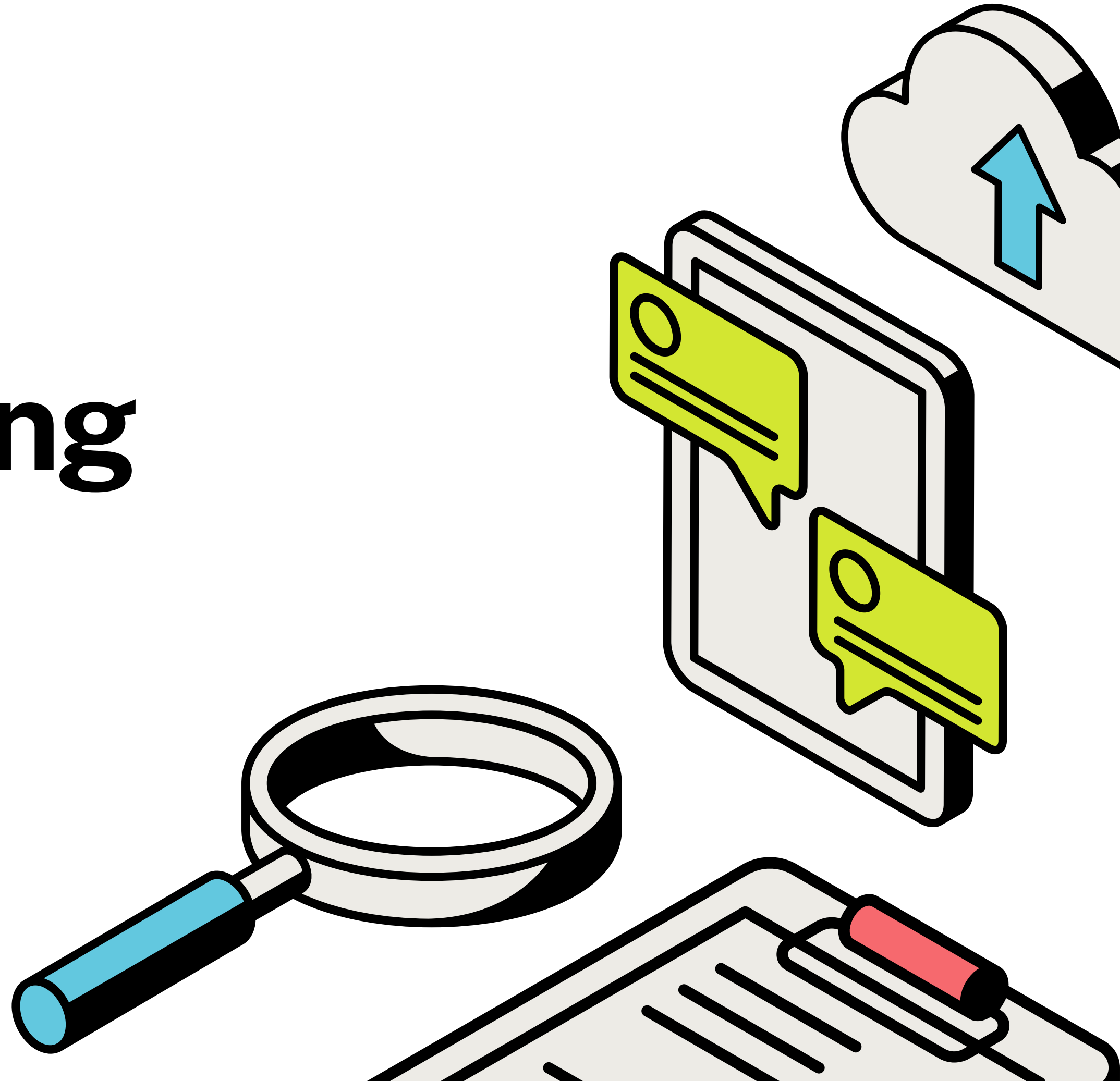




Fake News Detection using GNNs

Date: November 2025

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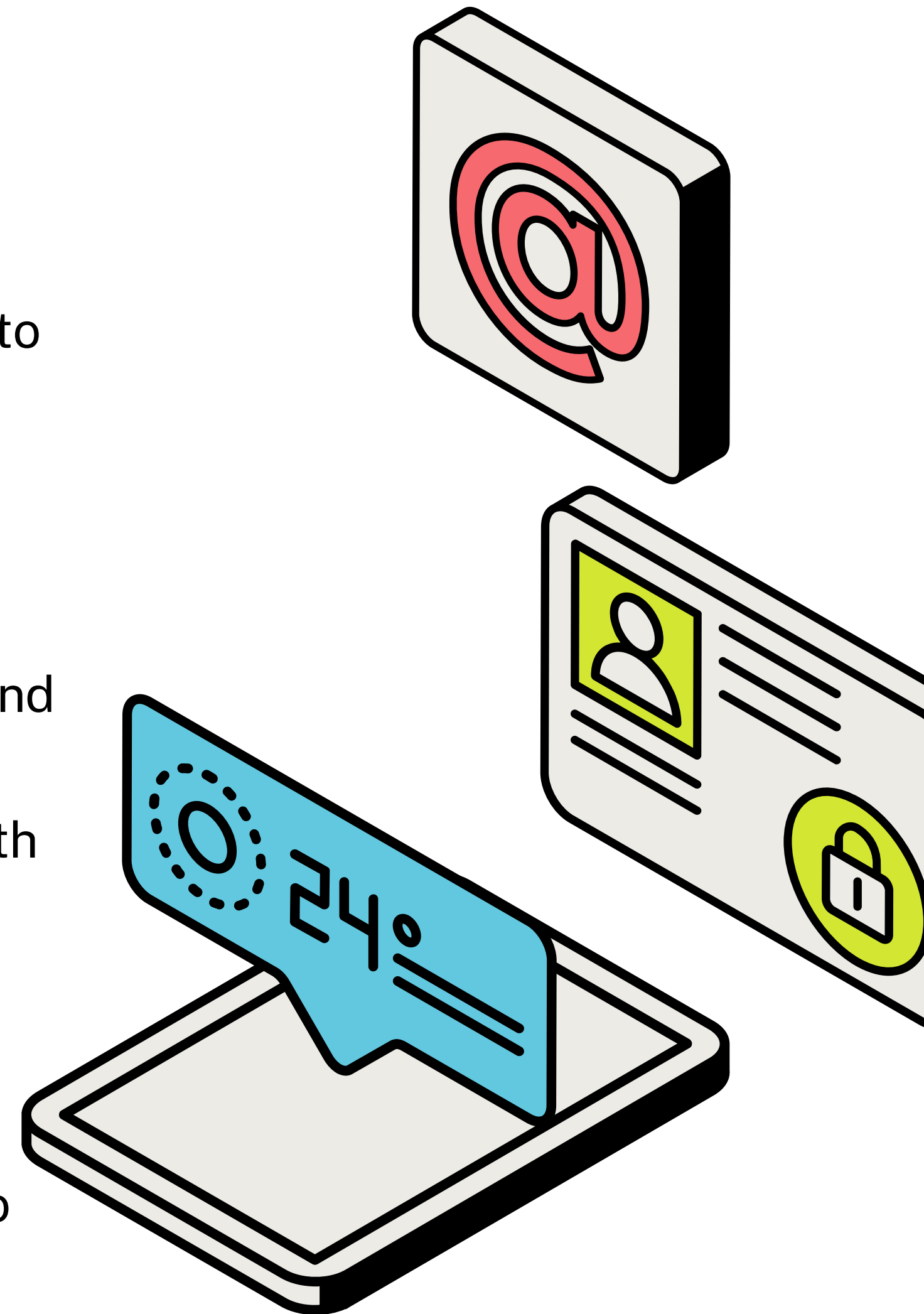


In this presentation:

1. Introduction & Definitions
2. Fake News features
3. Fake News Techniques
4. Taxonomies of GNNs
5. GNNs-based methods for FND
6. CGNs-based methods for FND
7. AGNNs-based methods for FND
8. GAE-based methods for FND
9. Conclusions and Open issues/challenges for future directions

Introduction

- Fake news is a source of absolutely inventive information to spread deceptive content or entirely misrepresent actual news articles.
- **Fake news include the following:**
 - Source: Publishers of news, such as authors, websites, and social networks.
 - Headline: Description of the main topic of the news with a short text to attract readers' attention.
 - Body content: Detailed description of the news, including highlights and publisher characteristics.
 - Image/Video: Part of the body content that provides a visual illustration to simplify the news content.
 - Links: Links to other news sources.



Why GNNs for FND?

- Fake news spreaders can attack machine learning-based models because these **models depend strongly on news text.** Making detection models less dependent on the news text is necessary to avoid this issue.
- GNN-based models can achieve similar or higher performance than modern methods **without textual information.**

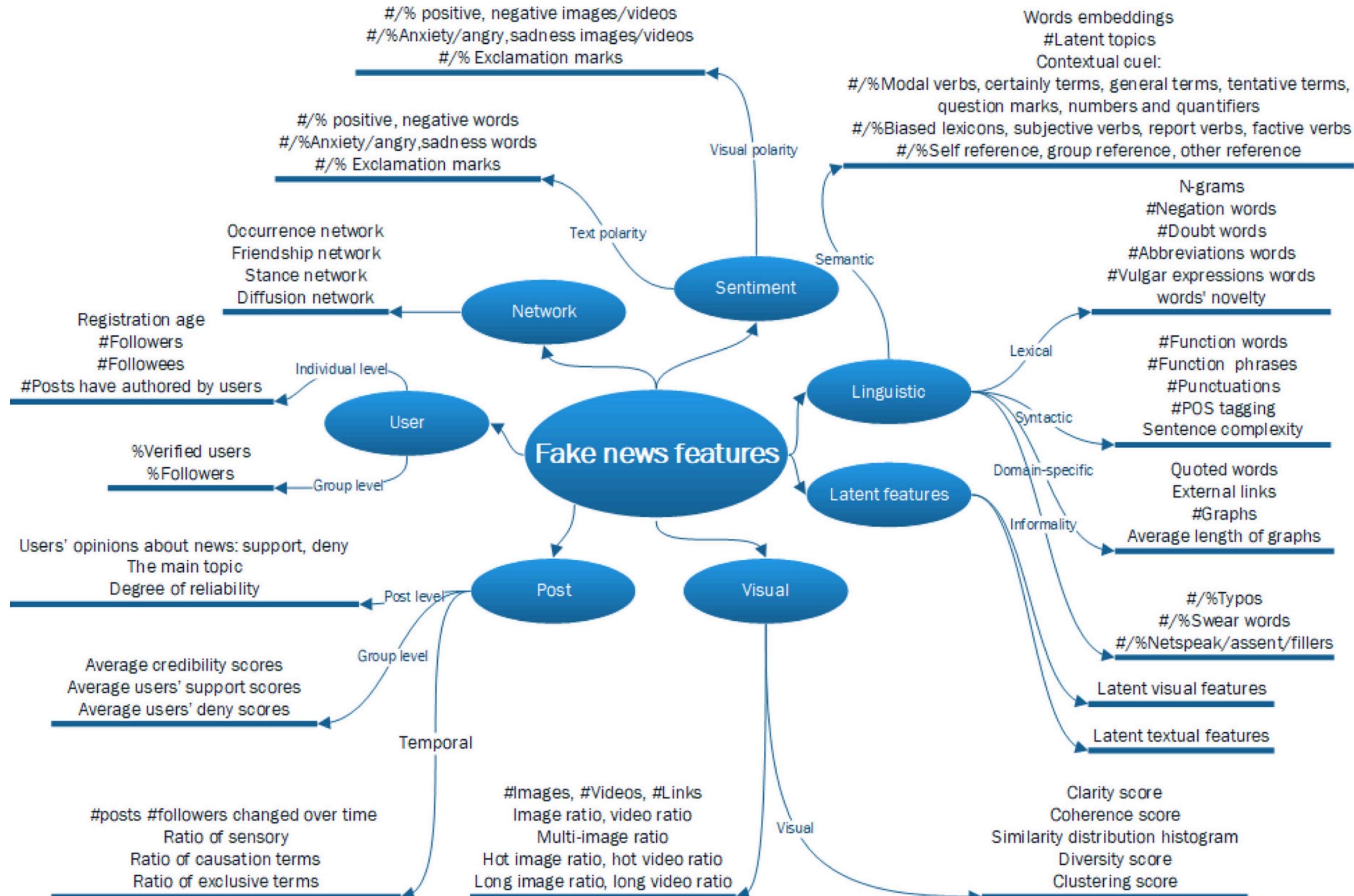
What's Fake News Detection (FND)?

- Given the social interactions ε among n users for news article a , the objective of fake news detection is to predict whether a is an instance of fake news.
- This objective is defined by a prediction function.

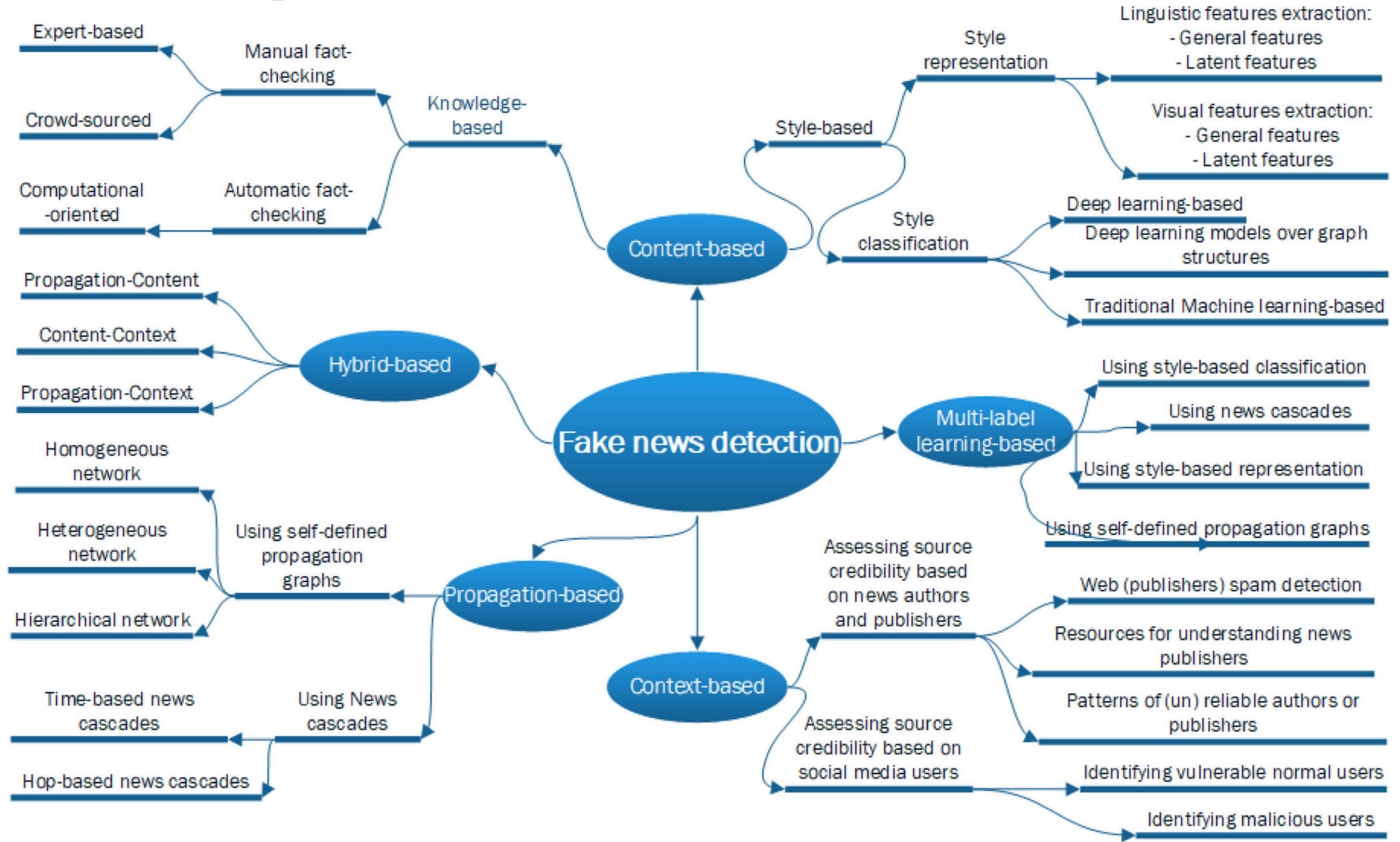
$F : \varepsilon \rightarrow \{0,1\}$ such that,

$F(a) = 1$, if a is a piece of fake news,
0, otherwise.

Fake News Features



FND techniques



Before we dive into GNNs for FND...

What is a graph?

- Formally, a simple graph is presented as $G=\{V,E\}$, where $V=\{v_1,v_2,...,v_n\}$ is the set of nodes, and $E=\{e_{11},e_{12},...,e_{nn}\}$ is the set of edges where $e_{ij}=(v_i,v_j)\in E$, $1\leq i,j\leq n$. In which, v_i and v_j are two adjacent nodes.
- The adjacency matrix A is a $n\times n$ matrix with $A_{ij} = \begin{cases} 1, & \text{if } e_{ij}\in E, \\ 0, & \text{if } e_{ij}\notin E. \end{cases}$

What Graph Neural Networks?

- Graph Neural Networks (GNNs) are a novel deep learning technique specifically designed to operate on *graph-structured data*. They generalize traditional neural networks to graphs by performing learning algorithms that aggregate and propagate information over nodes and edges in a graph.
- They come as useful in for ***modeling complex relationships in social networks***, where news, users, and interactions naturally form graph structures, ***overcoming the limitation*** of previous ML approaches that assumed news items were ***independent***.

Taxonomies of GNNs 1/4

1. Conventional GNNs:

- They came in as an extension of RNNs to graph structured data. Their mechanism is based on an information diffusion process where node states (embeddings) are iteratively updated by exchanging information with neighbours.

a) How GNNs Process Graph Data?

- GNNs process graph data by *passing information between nodes* through their connections (edges).
- Each node gathers information from its neighbours in a process called **message passing**, allowing it to learn representations based on its local structure and features.
- This iterative information-sharing allows nodes to understand the context provided by surrounding nodes in the graph.

Taxonomies of GNNs 2/4

b) A typical GNN operates in 3 steps:

1. **Initialization:** Each node is initialized with its feature vector, which could represent properties like age, gender, or molecular weight, depending on the application.
2. **Message Passing:** Over several iterations or "layers", nodes exchange information with their neighbours, aggregating data from connected nodes.
3. **Update:** Each node updates its feature vector using aggregated information, often by applying a neural network layer (e.g., a linear transformation followed by a non-linear activation function).

Taxonomies of GNNs 3/4

Graph Convolutional Networks (GCNs):

- They apply convolutional operations on graph data just like how it's done with images using traditional CNNs.
- The main idea is to generate a node's representation by aggregating its own features and neighbours' features.

Graph autoencoders (GAEs)

- They are unsupervised learning frameworks that encode nodes/graphs into a latent vector space and reconstruct graph data from the encoded information, as follows:
 - The encoder (often a GCN) produces latent node embeddings capturing both structural and content-related information.
 - The decoder reconstructs adjacency or attribute matrices to ensure meaningful

Taxonomies of GNNs 4/4

Spatial-temporal graph neural networks (STGNNs)

- They are designed for dynamic graphs where node features and graph structures vary over time.
- They consider spatial dependency and temporal dependency at the same time by combining spatial graph convolutions with temporal models such as RNNs or CNNs to capture temporal dependencies along with spatial graph relations.

Attention-based graph neural networks (AGNNs)

- AGNNs enhance standard GNNs by integrating attention mechanisms, which dynamically weigh the importance of neighbouring nodes during feature aggregation.
- This adaptive weighting allows AGNNs to focus more on influential nodes or edges that contribute significantly to the detection task.
- AGNNs naturally handle the heterogeneity and complexity inherent in social networks and fake news propagation.

Few of GNNs based FND approaches 1/2

1. *Continual Learning for Fake News Detection from Social*

- They exploit GNNs*' capability with non-Euclidean data (graph data in this case) to distinguish propagation patterns of real versus fake news on social media.
- Two GNN instances are trained: one on complete data and another on partial data using **continual learning** techniques (**Gradient Episodic Memory (GEM)** and **Elastic Weight Consolidation**)
- Those continual learning techniques enables **early fake news detection** by adapting to new data without the need to retrain on the whole dataset.

2. *Fake news detection in social media using graph neural networks and nlp techniques: A COVID-19 use-case*

- They focus on detecting malicious users spreading misinformation about COVID-19 and 5G networks conspiracy theories (tweets).
- They employ two strategies (*Content-based* fake news detection and *Context-based* fake news detection to learn representations and classify comments into categories (non-conspiracy, conspiracy, other conspiracies))

Few of GNNs based FND approaches 2/2

3. *FANG: leveraging social context for fake news detection using graph representation*

- FANG (Factual News Graph) is a *context-based model* that focuses on capturing news context, including sources, users, interactions and timelines.
- It constructs two ***homogeneous subgraphs***: one for *news sources* and one for *users*, both processed by an *unsupervised model* that captures neighbours relations.
- It also uses a ***pretrained detection network*** for news content extraction as *auxiliary inputs*.
- It achieves **robustness** even with *limited training data*.

Few of CGNs based FND approaches 1/2

1. Rumor Detection Based on SAGNN: Simplified Aggregation Graph Neural Networks

- This method aims to calculate the degree of interaction between twitter users for rumour detection.
- It implements a simplified version of GCNs by eliminating the weight matrices.
- It also adjusts the adjacency matrix to incorporate parent (source tweet) -child tweet (responses or retweets) relationships explicitly.

2. Detection of rumor conversations in Twitter using graph convolutional networks / Applied Intelligence

- *This method constructs two independent GCNs:
 - * GCN of tweets (source and reply; tweet i replies to tweet j)
 - * GCN of users (interactions among users; user i sent m tweets to user i in conversation)
- These two GCNs are concatenated into one fully connected layer for final classification.

Few of CGNs based FND approaches 2/2

3. ***GCAN: Graph-aware Co-Attention Networks for Explainable Fake News Detection on Social Media***

- This method integrates ***dual co-attention mechanisms*** with GCNs.
- The **first** simultaneously captures *relations between tweet content and user interactions*.
- The **second** simultaneously captures *relations between tweet content and user propagation*.
- This multi-level attention captures deep correlations between content and social context, helping to distinguish fake from real news effectively.

AGNNs based FND approach 1/2

Adversarial Active Learning Based Heterogeneous Graph Neural Network for Fake News Detection |AA-HGNN

- *This method constructs a **heterogeneous graph** or a Heterogeneous Information Network (HIN) comprising of different types of nodes (creators, news, subjects) and edges (write, belongs to).*
- *It employs a **two-level attention mechanism** defined below:*
 - 1. First Level: Attention at Neighbour Node Level:***
 - The model focuses on learning the importance (weights) of individual neighbours of a given node.*
 - For each neighbour node, the AGNN calculates an attention score that reflects how relevant or influential that neighbour is to the target node.*
 - 2. Second Level: Aggregating Type-Specific Neighbour Weights:***
 - The nodes have neighbours of different types (creators, news, subjects)*
 - The second attention level aggregates the attention scores learned from neighbours of the same type (called type-specific neighbours).*

AGNNs based FND approach 2/2

- *In addition to the attention mechanism implemented, **AA-HGNN combines adversarial learning and active learning.***
- AA-HGNN incorporates **active learning** where a **selector module** identifies high-value candidate nodes from the unlabelled set to be annotated and included in the training data.
- The selector is trained in an **adversarial way**; The selector and the classifier are competing against each other as follows:
 - The selector looks for the challenging/informative nodes that can "fool" the classifier.
 - The classifier tries to accurately classify both labelled nodes and those selected in an adversarial way.
- After active learning optimization, the classifier produces the **final fake news classifications** on the node embeddings enriched through attention mechanisms.

GAEs based FND approach

A Graph Convolutional Encoder and Decoder Model for Rumor Detection

- This method proposes a model that captures textual, propagation, and structural information from news for rumour detection.
- It includes 3 components:
 - * The encoder uses a GCN to represent news text to learn information, such as text content and propagation.
 - * The decoder uses the representations of the encoder to learn the overall news structure.
 - * Implemented simultaneously as the decoder, The detector also uses the representations of the encoder to predict whether events are rumours.

Discussions over approaches

The following comprises of conclusions deducted from the surveyed approaches

GNNs	To develop new GNN-based models in the future, more attention should be given to extracting excellent features and building good standard data instead of focusing on improving the graph structure.
GCNs	When building fake news detection models based on GNNs, different graphs should be constructed to capture each specific type of information and then perform the fusion step.
GAEs & AGNNs	The heterogeneous graph was evaluated as superior to the homogeneous graph because it can capture more meaningful information.

Challenges as possible future directions in FND

Down below are some of the challenges related to GNNs in FND that can be presented as possible future directions.

1	Deepfake currently poses a significant challenge to tackle in FND
2	Detecting whether the influencers' posts are real or fake is a challenge when they are exposed to possible hacking of their accounts to spread fake news or disinformation.
3	Real-time FND is yet to be addressed in an era where news could be fake at one point and real at another.
4	Constructing benchmark datasets and determining the standard feature sets corresponding to each approach for fake news detection remain challenges.
5	Most GNNs resort to undirected graphs and edge weights as binary values (0,1) which is unsuitable in many real-life cases. That's why future studies can construct graphs with the weights of edges as the actual values representing the relationship among the nodes as much as possible
6	For NLP tasks, GNNs have not represented node features by capturing the context of a paragraph or an entire sentence , and can also overlook semantic relationships among phrases in the sentences.
7	NO GNNs have simultaneously considered all the content, context, common sense knowledge and semantic relations which are deemed essential for GNN-based NLP tasks.

Conclusions & Open issues

GNN-based fake news detection is relatively new. Thus, the number of published studies is limited, yet shows promising results. Additionally, many challenges need to be addressed to achieve more comprehensive results.

Graph-based fake news detection benchmarks may present an opportunity and direction for future research.

The current GNN-based methods have a **static** structure, difficult to adapt in real time, which calls for the construction of **dynamic graphs** that are spatiotemporally capable of changing with real-time information.

Early detection of fake news involves detecting fake news at an early stage before it is widely disseminated so that people can intervene early, prevent it early, and limit its harm.

Majority of existing GNN-based methods work with homogeneous graphs. The use of **heterogeneous graphs** that contain different types of edges and nodes is thus a future research direction.

Absence of a **hybrid** of propagation, content, and context simultaneous usage in one model

Thank you.



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